



Team ID: 19
Section: 13



Members:
S VENUGOPAL (2520090185)
V PAVAN KUMAR (2520090079)
ABHISHEK YADAV (2520090208)



Guide:
Dr. M. Gayatri
Associate Professor



Abstract

The Wines Management System is a relational database system for managing wines, customers, suppliers, inventory, and sales orders. It uses MySQL, SQL, primary keys, foreign keys, CRUD operations, joins, and transactions to maintain accurate and organized data.



Tools & Technologies



Problem Statement & Objectives

Problem:

- Manual records can cause data redundancy, inconsistency and difficulty in retrieving information.



Objectives:

- Maintain accurate wine, supplier, customer, inventory and order information.
- Provide fast access to stock, order history, total sales and low-stock products.



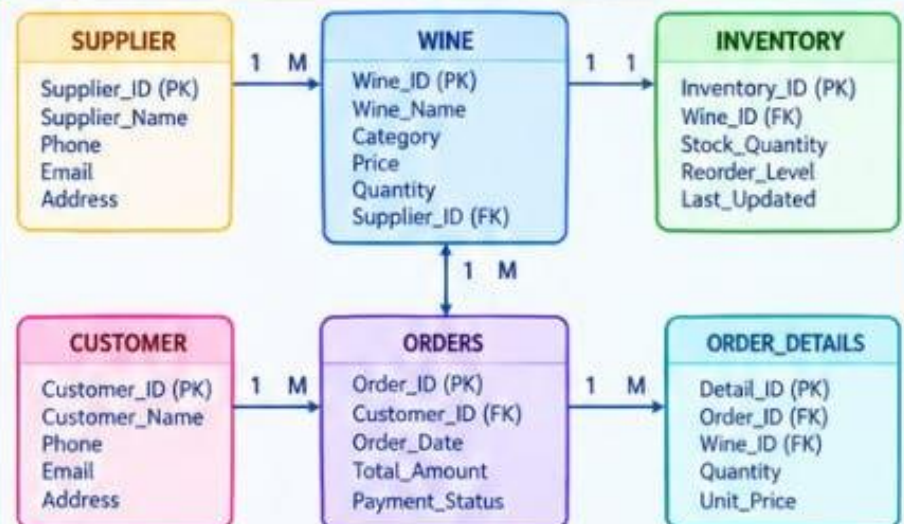
System Architecture



[Front-End – REST API – Database]



ER Diagram & Database Design



Existing vs Proposed System

Feature	Existing System	Proposed System
Data Storage	Manual / Spreadsheets	Centralized Database
Data Consistency	Often inconsistent	Maintains integrity with constraints
Retrieval Speed	Slow & difficult	Fast & easy
Relationship Mgmt	No proper links	PK-FK relationships
Reporting	Manual calculation	Automated (stock, sales, history, low stock)
Scalability	Limited	More modules / features



Feasibility Study

- Technical:** Required skills, tools and platform availability
- Economic:** Cost – open source stack, free-tier hosting
- Operational:** User acceptance, ease of use, maintenance
- Schedule:** Completion within the semester timeline



Results & Testing

- CRUD operations (Create, Read, Update, Delete)
- Data validation and constraints
- Joins, aggregate functions and transactions
- System performance – efficient and fast retrieval of data



Conclusion

- Achieved a centralized and efficient database system.
- Meets all project objectives: reduced redundancy, maintained consistency, organized access.



Future Scope

- Future scope includes a web interface, authentication, automated low-stock alerts, dashboards and online payment integration.



References

[1] Author, Title, Year [2] Website / Documentation link [3] Paper / Book



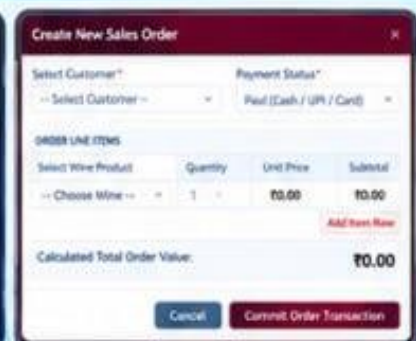
QR: GitHub



Implementation Screenshots



Dashboard



Create Order